# *Vegan diet-based lifestyle program rapidly lowers homocysteine levels.*

BACKGROUND: Plasma homocysteine levels have been directly associated with cardiac disease risk. Current research raises concerns as to whether comprehensive lifestyle approaches including a plant-based diet may interact with other known modulators of homocysteine levels. METHODS: We report our observations of homocysteine levels in 40 self-selected subjects who participated in a vegan diet-based lifestyle program. Each subject attended a residential lifestyle change program at the Lifestyle Center of America in Sulphur, Oklahoma and had fasting plasma total homocysteine measured on enrollment and then after 1 week of lifestyle intervention. The intervention included a vegan diet, moderate physical exercise, stress management and spirituality enhancement sessions, group support, and exclusion of tobacco, alcohol, and caffeine. B vitamin supplements known to reduce blood homocysteine levels were not provided. RESULTS: Subjects' mean homocysteine levels fell 13%: from 8.66 micromol/L (SD 2.7 micromol/L) to 7.53 micromol/L (SD 2.12 micromol/L; $P < 0.0001$). Subgroup analysis showed that homocysteine decreased across a range of demographic and diagnostic categories. Conclusions. Our results suggest that broad-based lifestyle interventions favorably impact homocysteine levels. Furthermore, analysis of Lifestyle Center of America program components suggests that other factors in addition to B vitamin intake may be involved in the observed homocysteine lowering.